

DS604, Introduction to Data Management (Autumn'2025)

Course Placement: M.Sc. (Data Science), Core, Semester-I

Course format: 3 Lecture hours and 2 hours Lab

Course Instructor: PM Jat, 2203, Faculty Block 2

Course Content:

Introduction to Data Management is designed to provide students with a foundational understanding databases that spans to

- (1) Data, databases, data models, data representations and operations on them. Roles of metadata, and database management systems
- (2) Relational Model as database implementation model, its concepts, key constructs, and manipulations
- (3) Manipulation Operations through Relational Algebra visa-a-vis SQL
- (4) ER Modeling: Common artifacts of ER Modeling, and their uses in depicting a database design in an implementation independent manner.
- (5) Formal approaches for Designing and evaluation of relational databases
- (6) Techniques to created database server hosted programs that run in the close proximity of the data providing efficient execution and provide highest level of shareability of the functionality
- (7) Technique for efficiently storing the data that provide efficient access for answering the queries
- (8) Concepts of Query Optimization and Execution for relational systems.
- (9) Concerns and Techniques to address those concerns while concurrent Transaction Processing and Technics that are used in dealing with concurrent

Labs and Assignments:

During Lab hours, students practice concepts learned in theory and would be asked to design and implement a database system for a mid-size real-life scenario.

Text Books:

1. **Fundamentals of Database Systems**, Ramez Elmasri and Shamkant B. Navathe, Pearson Education
2. **Database System Concepts** by Avi Silberschatz, Henry F. Korth, S. Sudarshan, TMH

Evaluation plan:

1. Two Mid Semester Exams: 30%
2. Project: 20%
3. Lab Evaluation: 20%
4. End Semester evaluation: 30%

Course Outcomes:

The database is one of the core competency areas for data science students. Through this course, students develop the ability to

- Design and implement Database systems.

- Write and solve complex queries systematically
- Write programs that access databases for further processing

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
x	x	x		x				x			x

Tentative Lecture Distribution

Sl. No.	Description		No. of Lectures
1	Introduction to Databases		3
	1.1	Database, Instance, Schema, and Metadata	1
	1.2	Conceptual and Implementation database Models	2
2	Relational Model and Operations on Relations (All operations are discussed in Relational Algebra and SQL side by side)		7
	2.1	Relational Concepts	1
	2.2	Basic Relational Operations (RA and SQL)	2
	2.3	Set Operations (RA and SQL)	1
	2.4	Aggregation Operations (RA and SQL)	1
	2.5	Division Operation (RA and SQL) and more SQL operations	2
3	Entity Relationship Modeling		6
	3.1	Concepts and notations: Entity/Relationship type, instance, and sets	1
	3.2	Aggregation, Ternary Relationships and case study	1
	3.3	ER to relational Mapping	1
	3.4	Enhanced ER Modeling and EER to relational mapping	1
	3.5	Case study discussion	2
4	Database Design		6
	4.1	Database Design Concepts and Design Anomalies in relations	1
	4.2	Theory of Functional Dependencies	2
	4.3	Normal Forms as measure of “goodness” of a relation design	1
	4.4	Decomposition Principles and Algorithms	2
5	Database Programming		4
	5.1	Views and Stored Procedures	2
	5.2	API based access of databases	2
6	Storage and Indexes		4
	6.1	Database storage file structures	2
	6.2	Indexing Methods and SQL support	2
7	Query Execution and Optimization		4
	7.1	Query Execution and Optimization Concepts	2
	7.2	Algorithms for basic operations	2
8	Overview of Transaction Processing		5
	8.1	Transaction Concepts	2
	8.2	Overview Concurrency Techniques	2
	8.2	SQL Isolation Levels	1
Total Lectures			39